

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
COLLEGE OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF ELECTRONICS AND INSTRUMENTATION ENGINEERING
Cycle Test –II

Academic Year: 2024 – 25

Program offered: M.Tech- ECE & AR

Course Title: AI in Automation

Course Code: : 21EIC505T

Duration: 1 period

Date of Exam: 11.12.2024

Year / Sem: I/I

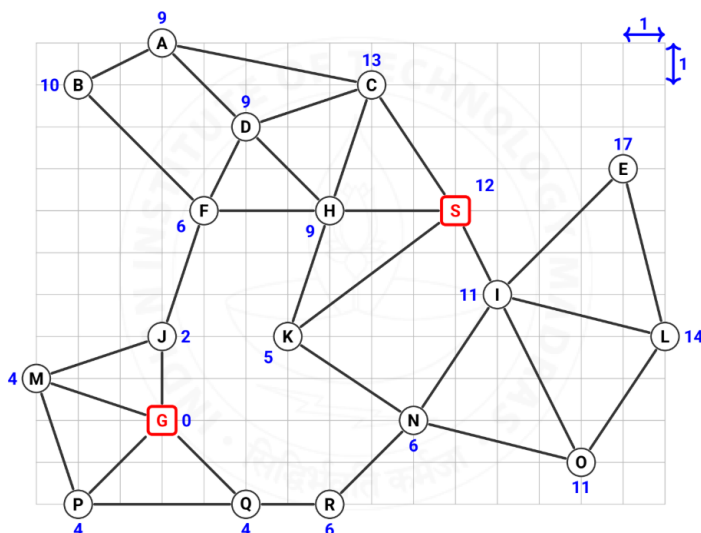
Max. Marks: 30

Answer any 3 questions

(3x10=30)

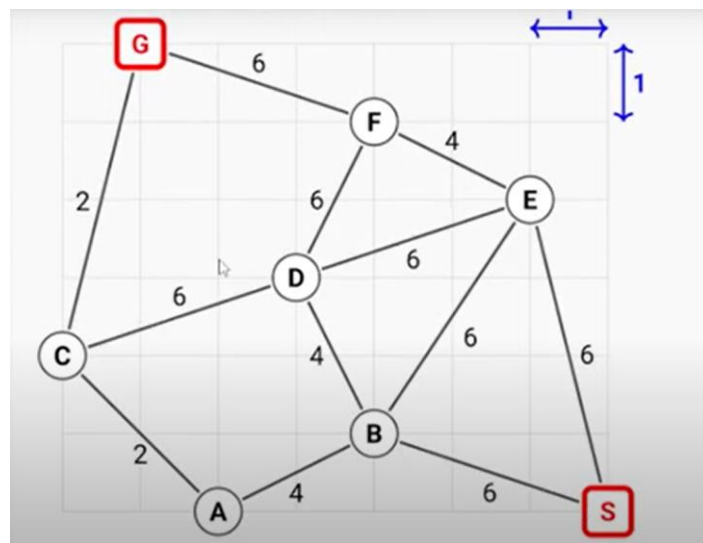
1.The figure shows a map with several locations on a grid where each tile is 1x1 in size. The locations are at grid points, and the locations are connected by two-way edges (roads). The start node is S and the goal node is G, the MoveGen function returns neighbours in alphabetical order. The RemoveSeen procedure removes neighbours already present in OPEN/CLOSED lists. Use Manhattan distance as the heuristic function and best first search to generate the state space.

State Space



(i) List the nodes returned by MoveGen(S). Enter the nodes in the order in which MoveGen returns them.

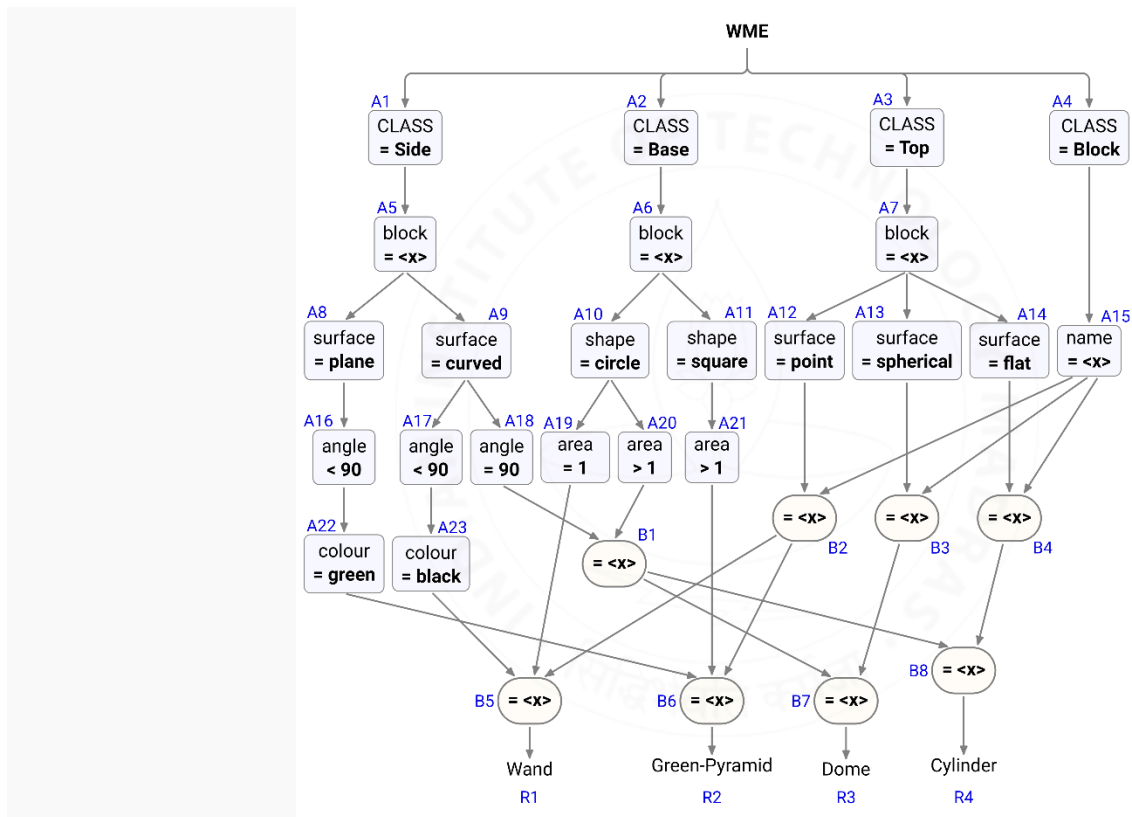
2. The figure shows a map with several locations on a grid where each tile is 1x1 in size. The locations are at grid points, and the locations are connected by two way edges (roads). The start node is S and the goal node is G, the MoveGen function returns neighbours in alphabetical order. The RemoveSeen procedure removes neighbours already present in OPEN/CLOSED lists. Use Manhattan distance as the heuristic function and Best first search to generate the state space.



(i) List the nodes returned by MoveGen(S). Enter the nodes in the order in which MoveGen returns them.

(ii) Compute the heuristic values of the nodes.

3. A variation of the Rete Net used in the lecture is shown in the figure. The labels A1, A2, A3, ..., A10, A11, A12, ..., B1, B2, ..., R1, R2, ..., uniquely identify the nodes in the network. Where required, use the above label ordering to break ties and to enter your answers.



Run the Rete algorithm for the Working Memory shown below, here the WMEs are in timestamp order. Assume that WMEs reside at appropriate Alpha nodes, and the Beta nodes point to WMEs residing in Alpha nodes.

101. (block ^name 1 ^owner Harry)
102. (block ^name 2 ^owner Hermione)
103. (block ^name 3 ^owner Ron)
104. (block ^name 4 ^owner Draco)
105. (base ^block 1 ^shape circle ^area 1)
106. (base ^block 2 ^shape square ^area 20)
107. (base ^block 3 ^shape circle ^area 10)
108. (base ^block 4 ^shape circle)
109. (side ^block 1 ^angle 80 ^surface curved ^color black)
110. (side ^block 2 ^angle 60 ^surface plane ^color green)
111. (side ^block 3 ^angle 90 ^surface curved)
112. (side ^block 4 ^angle 50 ^surface curved)
113. (top ^block 1 ^surface point)
114. (top ^block 3 ^surface flat)

For each WME, identify its location (node label) in the Rete Net, then prepare the conflict set for the first cycle, then answer the questions.

- (i) The location(s) of WME 101 is/are _____.

- (ii) List the WMEs residing at A15 _____. Identify their timestamps.

4. The figure shows a 4-ply game tree with evaluation function values at the horizon. The nodes in the horizon are assigned labels A,B,C,...,P. Use these labels when asked to enter a horizon node or a list of horizon nodes. The internal nodes are labeled in a level-by-level left-to-right manner:

Max root : a1

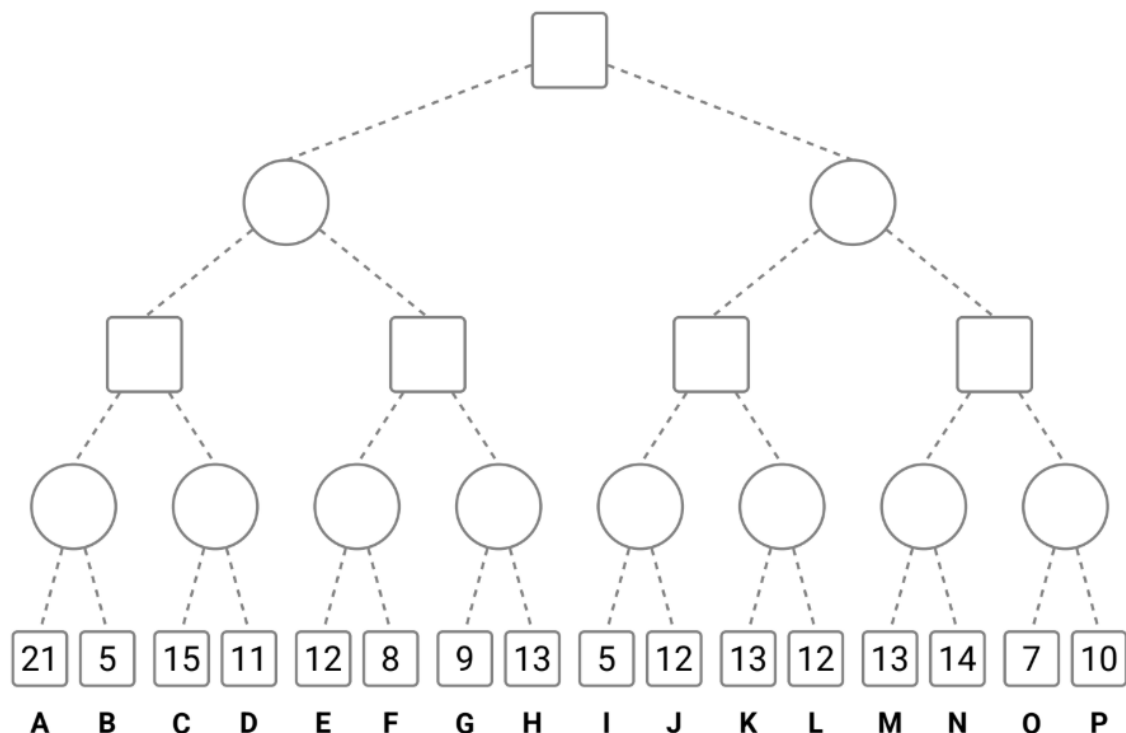
Min level: b1,b2

MAX level: c1,c2,...,c4

MIN level: d1,d2,...,d8

Horizon (MAX) level: A,B,C,...,P

Tie-breaker: when several nodes have the same h-value then select the deepest leftmost node.



- (i) What is the MinMax value of the game?
- (ii) List the horizon nodes pruned by alpha-cuts and beta-cuts.